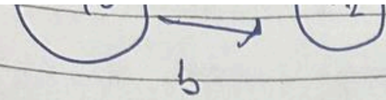
Nhean omra



Practice

1 finite Automata

a determine the definition of each component in machine $M = (Q; \Sigma; \delta; q_0; F)$

+ Q : finite ~~status~~ set of state
+ Σ : Set input symbols
+ δ = Transition function
+ q_0 = initial state
+ F = set of final states

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~~Practice~~

Determine the defination of each component in machine $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$\delta = \{$

$q_0 = q_1$ is the initial state

$F = q_2$

δ =

	0	1
q_1	q_1	q_2
q_2	q_3	q_2
q_3	q_2	q_2

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C sketch the DFA given : $M = (Q, \Sigma, \delta, q_0, F)$

where : $Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the initial state

$F = \{q_0, q_1, q_2\}$

δ is defined as follow

Initial state q	symbol σ	final state $\delta(q, \sigma)$
q_0	a	q_0
q_0	b	q_1
q_1	a	q_0
q_1	b	q_2
q_2	a	q_0
q_2	b	q_2
q_3	a	q_3
q_3	b	q_3

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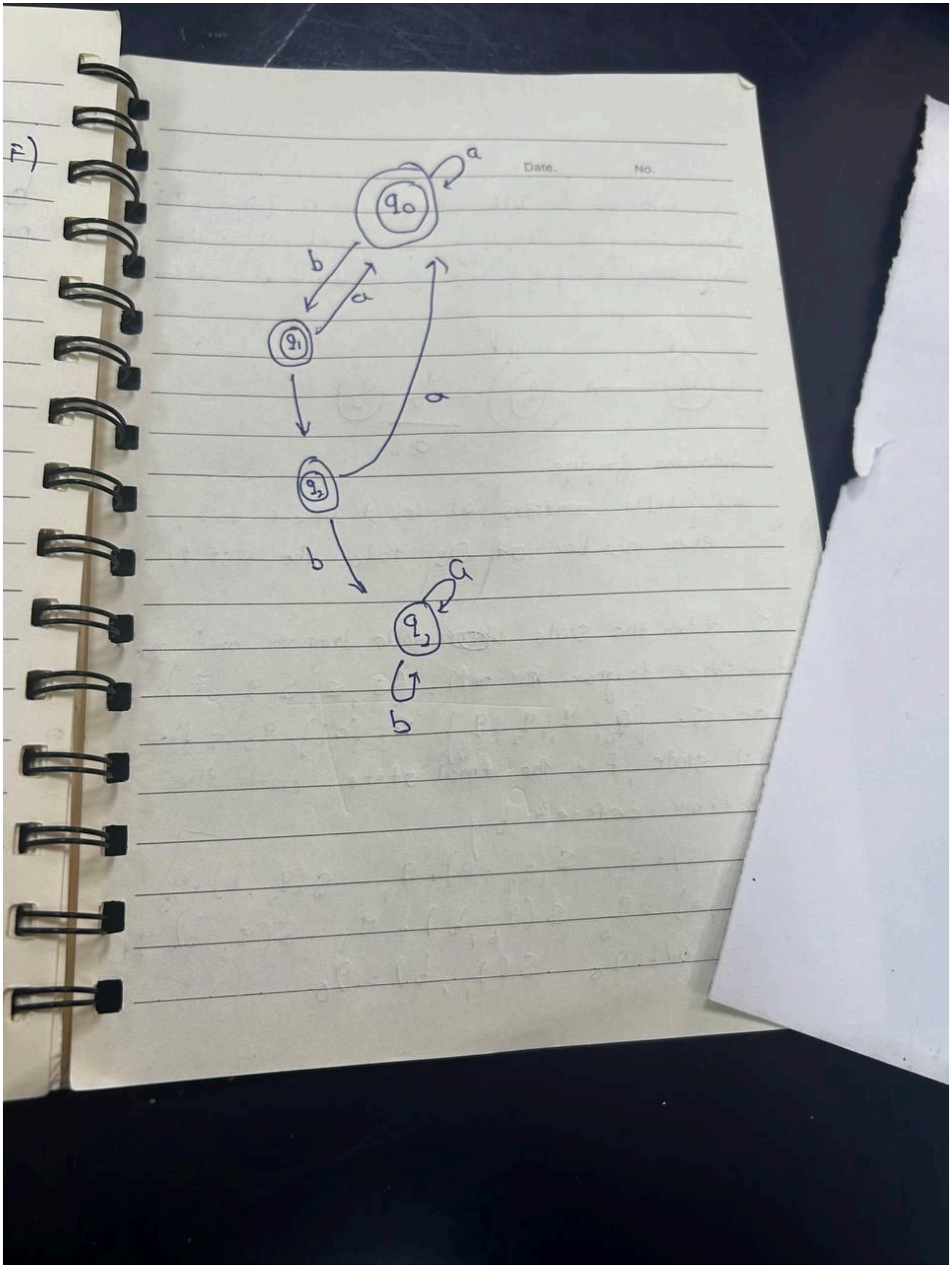

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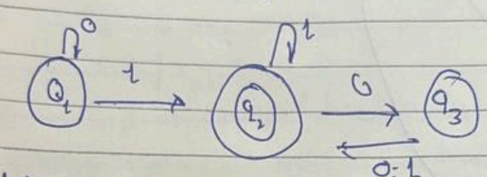
d determine the DFA schematic for $M = (Q, \Sigma, \delta, q, F)$ where $Q = \{q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$, q_1 is the start state, $F = \{q_2\}$ and δ is define as follows



determine longuge

$A = \{ w \mid w$ contains at least one 1 and even number of 0s follow the last 1 $\}$

e obtain the state ~~where~~ table diagram and state transition diagram for $M = (Q, \Sigma, \delta, q_0, F)$ where $Q = \{q_0, q_1, q_2, q_3\}$, $\Sigma = \{a, b\}$, q_0 is the initial state, F is the final state with the trasition define by

$\delta(q_0, a) = q_2$ $\delta(q_3, a) = q_1$ $\delta(q_2, b) = q_3$

$\delta(q_1, a) = q_3$ $\delta(q_0, b) = q_1$ $\delta(q_3, b) = q_2$

$\delta(q_2, a) = q_0$ $\delta(q_1, b) = q_0$

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$Q = \{q_0; q_1; q_2; q_3\}$

	a	b
q_0	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2